DSP- Revision Sheet # 2 (CH- 3 and CH- 4)
Summer 26

Chapter # 3:

1- Define the following items: Correlation operation, Correlation coefficient, Convolution operation and end effect problem and its solution.
2- Describe the convolution operation and deduce its compact relations.
3- Using direct method, calculate the cross-correlation of the following data sequences, $x_1 = \{1, 1, 0, 1\}$ and $x_2 = \{1, 0, 0, 1\}$ and plot the obtained correlation function against lag number, j.
4- For the tow sequences given in Prob.3,{1, 1, 0, 1} and {1, 0, 0, 1}", calculate their linear cross-correlation and plot the correlation function against lag number, j.
5- Given the two sets of data {1.5, 2.0, 1.5, 2.0, 2.5} and {0, 0.33, 0.67, 1.0} calculate their Cross-correlation using direct method.
6- Examples 3.3, 3.5 and 3.6.

Chapter # 4:

1- For LTIS define the following items: Linearity, Causality Stability, ROC, difference Eq. and BIBO Condition.
2- Power series, partial fraction and residue method problems for finding the inverse Z Transform.
3- Examples 4.1, 2, 3, 4, 6, 7, 9 and 10.

Chapter # 5:

1- What are the steps of digital filter design?
2- Compare between FIR and IIF filters.
3- Mention four from the digital filter advantages.
4- two problems from that given in DSP section, for filter design.
5- Problems attached with Lecture report (digital filter report).